DESIGN REPORT

EMPLOYEE COMPENSATION MANAGEMENT SYSTEM

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1. Assumptions

The problem statement is open, so these are the decisions I made:

- Currency is VND, stored as `Double` and formatted by a `vnd` helper.
- `id` and `hireDate` are immutable (`let`) — employee identity never changes.
- `performanceRating` is a 0…5 score, set by HR, and used to compute the performance bonus.
- There are three employment arrangements: full-time, part-time, and contract. Contract has two sub-models: fixed monthly or milestone-based.
- Three user roles exist: `hrStaff`, `payrollAdmin`, `viewer`. Each role declares which actions it may perform via a permission check inside `Employee.authorize(_:_:action:)`.
- Payroll is monthly. `PayMonth` represents one calendar month and can be stepped via `.previous` / `.next`.
- Once a month's payment for an employee is approved, the payment record becomes locked. The only allowed change is a correction (`submitCorrection`) that appends a `Correction` with a written reason — the approved amount itself is never modified.
- No backend — `PayrollManager(sample: true)` loads six employees and a couple of previous-month approvals; `PayrollManager.preview()` reuses it for SwiftUI previews.

2. Type identification

Classes:

Class	Responsibility
`Employee`	Base class. Holds shared identity (`id`, `hireDate`, `name`, `department`, `position`, `performanceRating`) plus the immutable payroll ledger (`paymentRecords`, `correctionLog`). Owns the payroll lifecycle: `approvePayment`, `submitCorrection`, `finalPayable`, and the `authorize` / `ensureMonthOpen` guards.

FullTimeEmployee	Fixed monthly salary + optional allowance + rating-based bonus.
PartTimeEmployee	Hourly rate × recorded work hours + rating-based bonus. Enforces hour limits.
ContractEmployee	Two contract models — fixed monthly payment or milestone-based. Manages milestone creation, amount editing, and approval.
PayrollManager	Owns all employees. Provides search, add, month summaries, approve-all, sample data.

Structs:

Struct	Fields	Responsibility
Milestone	`let id = UUID()`, `fileprivate(set) var title: String`, `fileprivate(set) var amount: Double`, `fileprivate(set) var approvedMonth: PayMonth?`, computed `isApproved`	One milestone under a contract. `approvedMonth` locks the milestone once paid.
WorkLog	`let id = UUID()`, `let date: Date`, `let hours: Double`	One hours-entry for a part-time employee.

Supporting types:

- `ContractModel` (enum) — fixed monthly / milestone-based
- `EmploymentType` (enum) — fullTime / partTime / contract
- `UserRole` (enum) — hrStaff / payrollAdmin / viewer, with `.can(_:)` permission check
- `Permission` (enum) — `editInfo`, `editCompensation`, `approvePayment`
- `PayMonth` (struct) — month identity with `.previous` / `.next` / `.contains(date)` / `.date(day:)`
- `PayLine` (struct) — one row on a payslip (label, amount)
- `PaymentRecord` (struct) — a frozen approved payslip
- `Correction` (struct) — post-approval adjustment (month, adjustment, reason, date)
- `PaySummary` (struct) — display model for the Payroll tab
- `InfoItem` (struct) — generic key-value row
- `PayrollError` (enum) — every validation error with a clear message

3. Properties & Methods

Shared stored properties in `Employee`:

- `let id`, `let hireDate` — immutable
- `private(set) var name`, `department`, `position`, `performanceRating`
- `private(set) var paymentRecords: [PaymentRecord]`
- `private(set) var correctionLog: [Correction]`

Overridable policy hooks in `Employee`:

- `employmentType`, `typeName`, `icon`, `bonusPolicyText`, `searchableText`, `compensationRows`
- `performanceBonusRate(for rating:) -> Double`
- `compensationLines(for month:) -> [PayLine]` — the main polymorphic method

Shared `final` methods (cannot be overridden by subclasses):

- `bonusLine(base:)`, `grossPay(for:)`, `displayLines(for:)`
- `record(for:)`, `isApproved(_:)`, `corrections(for:)`, `correctionsTotal(for:)`, `finalPayable(for:)`
- `authorize(_:_:action:)`, `ensureMonthOpen(_:)`
- `updateInfo(name:department:position:by:)`, `setPerformanceRating(_:by:)`
- `approvePayment(for:by:)`, `submitCorrection(for:adjustment:reason:by:)`, `overwriteApprovedPayment(for:newAmount:by:)` (always throws)

Subclass-specific stored properties (access level noted):

Class	Property	Access
`FullTimeEmployee`	`monthlySalary`, `allowance`	`private(set) var`
`PartTimeEmployee`	`hourlyRate`, `workLogs`	`private(set) var`
`ContractEmployee`	`contractModel`	`let`
`ContractEmployee`	`fixedMonthlyAmount`, `milestones`	`private(set) var`

Subclass-specific methods:

- `FullTimeEmployee.updateCompensation(salary:allowance:by:)`
- `PartTimeEmployee.recordHours(_:on:by:)`, `setHourlyRate(_:by:)`, `hours(in:)`, `logs(in:)`
- `ContractEmployee.setFixedAmount(_:by:)`, `addMilestone(title:amount:by:)`, `updateMilestoneAmount(id:newAmount:by:)`, `approveMilestone(id:in:by:)`, `approvedMilestones(in:)`

`searchableText` is declared on `Employee` as an extension point but is not used by the current search (which uses id/name/department/position). This is a deliberate placeholder.

4. Inheritance & Polymorphism

Hierarchy:

```
Employee
├── FullTimeEmployee
├── PartTimeEmployee
└── ContractEmployee
```

All three subclasses are declared `final`, so no ad-hoc extension below them. Every shared behavior above is `final`, so subclasses can only override the policy hooks — this keeps payroll math consistent across types.

Inherited (written once): identity fields, `paymentRecords` ledger, `correctionLog`, the approve/correct workflow, `authorize` and `ensureMonthOpen` guards, `bonusLine`, `grossPay`, `finalPayable`, `displayLines`.

Complete override matrix:

Member	FullTimeEmployee	PartTimeEmployee	ContractEmployee
employmentType	.fullTime	.partTime	.contract
typeName	"Full-time"	"Part-time"	"Contract (Fixed payment)" / "(Milestone-based)"
icon	briefcase.fill	clock.fill	doc.text.fill
bonusPolicyText	rating-based text	rating-based text	model-dependent text
performanceBonusRate	20% if ≥ 4.5, 10% if ≥ 4.0, else 0	10% if ≥ 4.5, else 0	5% if milestone-based and ≥ 4.5, else 0
compensationLines(for:)	salary + allowance + bonus	hours × rate + bonus	fixed monthly amount or approved milestones + bonus
compensationRows	salary / allowance / policy	hourly rate / monthly limit / policy	contract model / fixed amount or milestone count / policy

`detailRows` in `Employee` is not overridden — it provides the shared identity rows (ID, type, department, position, hire date, rating). Subclasses contribute extra display info

via `compensationRows` instead. This keeps the base class the single source of identity text.

Polymorphism in practice:

- `PayrollManager.summaries(for:)` calls `summary(of:month:)` for each `Employee`. Inside, `displayLines(for:)` resolves to the approved record or to `compensationLines(for:month:)`, which the actual type overrides. The manager never checks `is FullTimeEmployee` — each employee contributes its own lines.
- `finalPayable(for:)` sums the frozen record plus corrections. The base class does not care what type of employee produced the record.
- `EmployeeDetailView` does cast to specific types, but only to render the type-specific action forms (update salary, record hours, add milestone). The read-only payslip rendering is fully polymorphic.

When protocol/composition would be better: Swift cannot express "this employee is simultaneously full-time and contract-based". If the company later introduces hybrid arrangements, or an employee can hold multiple contracts at once, a class tree becomes awkward. In that case, protocols like `Payable`, `BonusEligible`, `MilestoneBased` and a `Contract` component would be cleaner. I chose class inheritance here because the three current types share one unified payroll lifecycle (approve, correct, lock) and the differences are entirely within `compensationLines` + `performanceBonusRate`.

5. Encapsulation & Validation

Access-control decisions:

Data	Access modifier	Reason
`Employee.id`, `hireDate`	`let`	Identity — must never change
`Employee.name`, `department`, `position`, `performanceRating`	`private(set) var`	HR must update them, but only via `updateInfo` / `setPerformanceRating` with validation and role check
`Employee.paymentRecords`, `correctionLog`	`private(set) var`	Ledger — changes only via `approvePayment` / `submitCorrection`
`FullTimeEmployee.monthlySalary`, `allowance`	`private(set) var`	Only `updateCompensation` may write
`PartTimeEmployee.hourlyRate`, `workLogs`	`private(set) var`	Only `setHourlyRate` / `recordHours` may write

`ContractEmployee.contractModel`	`let`	Fixed at creation
`ContractEmployee.fixedMonthlyAmount`, `milestones`	`private(set) var`	Only `setFixedAmount` / `addMilestone` / `updateMilestoneAmount` / `approveMilestone` may write
`Milestone.title`, `amount`, `approvedMonth`	`fileprivate(set) var`	Settable only inside `ContractEmployee.swift` — external code cannot change a milestone, and once `approvedMonth` is set the milestone is locked by the subclass
`PayrollManager.employees`	`@Published private(set)`	Views read the collection; only the manager appends
`PayrollManager.currentRole`, `selectedMonth`	`@Published` (writable)	The user picks these from the UI

Single gate for sensitive state: All mutating methods (`updateInfo`, `setPerformanceRating`, `approvePayment`, `submitCorrection`, `recordHours`, `updateCompensation`, `setHourlyRate`, `setFixedAmount`, `addMilestone`, `updateMilestoneAmount`, `approveMilestone`) are `final` in `Employee` or live only on a subclass. `overwriteApprovedPayment` is deliberately implemented to always throw — either `approvedPaymentLocked` (if approved) or `monthNotApproved`. This is the explicit rejection of "unauthorized change to an approved payment" required by Task 4.

`ensureMonthOpen(_:)` is a shared guard called from three places: `approvePayment` (before creating a record), `PartTimeEmployee.recordHours` (before appending a work log), and `ContractEmployee.approveMilestone` (before setting `approvedMonth`). This is what prevents any state change once a month is approved — the rule lives in one function and every sensitive write goes through it.

Validation rules:

Rule	Where	Error
Name/department/position not empty (after trim)	`updateInfo`	`emptyField`
Rating in 0…5, finite	`setPerformanceRating`	`invalidRating`

Hours finite, ≥ 0, ≤ 24 per entry	`PartTimeEmployee.recordHours`	`invalidHours`
Monthly total ≤ 120	`PartTimeEmployee.recordHours`	`monthlyHoursExceeded`
Month must not be approved before adding hours	`recordHours` (via `ensureMonthOpen`)	`monthAlreadyApproved`
Salary > 0, allowance ≥ 0, both finite	`FullTimeEmployee.updateCompensation`	`invalidAmount`
Hourly rate > 0, finite	`PartTimeEmployee.setHourlyRate`	`invalidAmount`
Contract model must be fixed-monthly	`ContractEmployee.setFixedAmount`	`notFixedContract`
Contract model must be milestone-based	`addMilestone`, `approveMilestone`	`notMilestoneContract`
Milestone title not empty, amount > 0	`addMilestone`	`emptyField` / `invalidAmount`
Milestone must exist	`updateMilestoneAmount`, `approveMilestone`	`milestoneNotFound`
Approved milestone cannot be edited	`updateMilestoneAmount`	`milestoneLocked`
Approved milestone cannot be re-approved	`approveMilestone`	`milestoneAlreadyApproved`
Milestone cannot be approved in a locked month	`approveMilestone` (via `ensureMonthOpen`)	`monthAlreadyApproved`
No duplicate employee ID	`PayrollManager.addEmployee`	`duplicateEmployee`
Month must be open to approve	`approvePayment` (via `ensureMonthOpen`)	`monthAlreadyApproved`

Total must be > 0	`approvePayment`	`nothingToPay`
Correction requires a written reason	`submitCorrection`	`correctionReasonRequired`
Correction must be non-zero and keep final ≥ 0	`submitCorrection`	`invalidCorrection`
Correction requires the month to be approved	`submitCorrection`	`monthNotApproved`
Direct overwrite of approved payment	`overwriteApprovedPayment`	`approvedPaymentLocked`
Role must have the required permission	`authorize`	`unauthorized`

6. SwiftUI implementation

Screen structure:

- `ContentView` — `TabView` with Employees and Payroll. Injects `PayrollManager(sample: true)` as `@StateObject`.
- `EmployeesView` — search field, segmented filter by `EmploymentType`, `RoleBanner` (current role + selected month), and a list of `EmployeeCard`. Cards show type color, name, department, current month's payable amount and Pending/Approved status.
- `EmployeeDetailView` — three stacked sections:
  - Information (shared `detailRows`) + Edit button.
  - Compensation details (polymorphic `compensationRows`) — automatically shows different rows for full-time, part-time, or contract.
  - Type-specific action form (`FullTimeSection` / `PartTimeSection` / `ContractSection`) — cast for the action form only; the payslip below is rendered polymorphically.
  - Monthly pay (`PaymentSection`) — month switcher, per-line amounts, corrections, net payable, an Approve button, an "Edit approved amount directly" button (always rejected), and a correction form.
- `PayrollView` — month switcher, total payable banner, "Approve all pending" button, and a list of `PayslipCard` for every employee.
- `EditEmployeeView` — sheet for updating name/department/position/rating. ID is shown as a locked `LabeledContent`, matching the `let` immutability in code.

Where polymorphic behavior is visible:

- Each employee card shows its own net payable for the selected month. Full-time shows salary + allowance + bonus; part-time shows hours × rate; contract shows fixed amount or approved milestones.
- The Compensation details section renders different rows per type without any `switch` in the view — it just iterates `compensationRows`.
- Bonus policy text differs per type (`bonusPolicyText`), matching `performanceBonusRate`.
- Milestone-only controls (Approve, Add milestone) appear only on `ContractEmployee` detail — driven by the type cast for the action form, not by the payslip renderer.

7. Conclusion

The design keeps shared payroll behavior (identity, approve, correct, lock, net calculation) in `Employee` and pushes only pay-rule variance into subclasses via `compensationLines`, `performanceBonusRate`, and `compensationRows`. Sensitive data (`id`, `hireDate`, `paymentRecords`, `correctionLog`, salary fields, `milestones`) is guarded with `let` or `private(set)`/`fileprivate(set)`, and every mutating operation goes through a `final` gate that first calls `authorize(role, permission, action:)` and then validates input. Approved payments can only be adjusted through the correction procedure — direct overwrite is explicitly rejected by design. The result is a compact model that is easy to extend (a new arrangement is a new subclass with three overrides) and easy to demonstrate in SwiftUI with clear valid/invalid feedback.